

EXACT 2026 Technical Solution Description

Team: AI WITH BRO | **System:** Local FastAPI service with sandboxed program execution and model orchestration

The system runs a local FastAPI service that maps educational queries to formal programs. Rather than relying on LLMs for direct calculation, the service serves as an orchestrator: language models generate code, an isolated sandbox executes it with Z3 and SymPy, and the final response schema is validated locally.

Architecture and Routing

The service exposes the official unified `/predict` endpoint. It accepts `query_id`, `type`, `query`, `premises`, and `options`, then routes deterministically by `type`. Legacy field aliases remain supported.

System Execution Pipeline

Stage	System Implementation	Engineering Control
Routing	Normalize official <code>query_id</code> , <code>type</code> , <code>query</code> , <code>premises</code> , and <code>options</code> fields	Deterministic fast routing.
Released-example retrieval	Exact normalized full-input match over disclosed Type 1 records	Returns the original 0-based premise indices; unseen inputs continue to reasoning.
Fast formulas	Solve strict common Physics patterns directly	SI conversion, series/parallel handling, and ASCII units avoid timeout.
Generator	Qwen2.5-Coder-7B; Qwen2.5-7B for direct choice/number/text answers	One self-hosted LLM resident at a time.
Sandbox	Subprocess script execution (Z3/SymPy)	AST constraint parsing, timeout daemon, isolated runtime.
Finalizer	Preserve verified solver output and evidence	Returns answer, explanation, FOL/CoT, premises, confidence.

Sandbox Security and Reliability

The execution sandbox protects the host system and ensures output validation:

- **AST Parsing:** Validates the compiled syntax tree of generated code. Restricts import modules strictly to `z3` and `sympy`, blocking standard runtime libraries (e.g., `os`, `sys`, `socket`).
- **Resource Caps:** Limits solver subprocess memory to 1536MB and locks execution time to a 20-second hard threshold.
- **Auto-Retry Loop:** Catches execution tracebacks and passes them back to the generator for a single correction try.
- **Budget Handler:** Tracks the global request duration. If elapsed time exceeds 58 seconds, it bypasses subsequent LLM nodes and serializes the raw sandbox trace to compile a compliant fallback response.
- **Runtime Retrieval:** No Physics vector index is shipped or active; the Physics RAG node safely no-ops unless a disclosed index is explicitly built.

Model Orchestration and Serving

Local inference uses `llama.cpp` and `llama-server`. The custom `LlamaServerSupervisor` manages model transitions:

- **Active Lock:** Ensures only one model runs in GPU memory to prevent VRAM allocation crashes.
- **Sequential Serving:** Swaps roles only between stages; parallel LLM inference is never used.
- **Protocol:** Exposes a self-hosted API compatible with standard SDK structures, running completely offline.